

Oleksii Bilozero

Senior Frontend Engineer | Frontend Lead

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Summary

Frontend engineer with 5+ years of experience building complex TypeScript products, including ~3 years as the sole architect and de-facto frontend lead of Embedd's VS Code extension for AI-powered embedded development, used by automotive and aerospace clients. I own a product frontend end-to-end – from monorepo architecture and build tooling to LLM agent integration, visual editors, and real-time hardware communication – and ship it continuously to the VS Code Marketplace.

Technical Skills

- **Languages & Runtime:** TypeScript, JavaScript, Node.js, HTML5, CSS3
- **Architecture:** monorepo & micro-frontend architecture (npm workspace, Turborepo, Vite, webpack), event-driven architecture, component-based design, SOLID
- **Frameworks:** React, Next.js, Remix.js
- **AI & real-time:** LLM agent integration (tool-calling protocols, streaming chat), WebSocket (Socket.IO, Rails ActionCable), Web Workers, Protocol Buffers
- **UI Libraries:** Tailwind CSS, Material UI, Headless UI, Chakra UI, Styled Components, Atlassian UI, CSS Modules
- **APIs & Integration:** VS Code Extension API, Chrome Extension API, Web Serial / Web USB API, Canvas
- **Practices:** strict TypeScript standards, code review, CI type-check gating, conventional commits & automated changelogs

Work Experience

Senior Frontend Engineer (Frontend Lead) — Embedd (Product)

January 2024 – Present

Embedded electronics development automation

Own the frontend of Embedd's flagship product end-to-end: a cross-platform VS Code extension for AI-powered embedd development and testing, serving major

microelectronics manufacturing clients including Bosch, Renesas, Microchip.
Sole architect and author of 100% of the frontend codebase; define the architecture, coding standards, and release process for the product.

- Architected a TypeScript monorepo of **16 webview micro-frontend apps and 8 shared packages** (npm workspaces, Turborepo, TypeScript project references) with dual build targets – native VS Code and self-hosted VS Code (web)
- Built a **visual device-tree editor for Zephyr RTOS** on a custom-forked rete.js node-graph engine: drag-and-drop SoC/peripheral configuration, subsystem-aware layout, cross-protocol bus-binding rules, undo/redo, and a persistence model that round-trips into generated device-tree source code.
- Integrated an **LLM agent into the editor**: a dedicated AI chat panel communicating with an agent harness over Socket.IO, implementing the client side of a tool-calling protocol (agent-invoked frontend tools with zod-validated contracts) that lets the agent inspect and modify the user's hardware configuration; shipped file attachments, session restore, and feature-flagged model selection.
- Built a **real-time firmware validation pipeline** over Rails ActionCable WebSockets (custom cable client, request/response correlation, transport-agnostic service layer), complemented by an in-editor Web Worker validator for instant feedback.
- Implemented **live communication with physical hardware**: Web Serial / SerialPort integration and a Protocol Buffers–based live-debug protocol, with time-series chart visualization for test data and device monitoring.
- Created a **firmware emulator sandbox** (React) for scenario-based testing with error-code diagnostics, running both standalone and embedded in the extension.
- Contributed **Device Tree Source language support** to the extension (TextMate grammar, language configuration) and designed the product onboarding (walkthroughs, activity-bar explorer).
- Established engineering standards across the codebase — strict TypeScript with no unsafe casts, type-guard narrowing, a typecheck-gated build graph — and reviewed contributions from other engineers. [adjust if you formally mentored anyone]
- Collaborated directly with backend and embedded systems engineers to design cross-service contracts (agent tool schemas, validation channels) and translate complex hardware workflows into intuitive UI.

Fullstack Engineer— Gearheart.io (Outsource)

May 2023 – November 2023

Outsourcing company • Internal tools

Developed full-stack solutions for internal dashboard applications, working across both backend and frontend technologies to deliver business-critical tools for client management and project tracking.

Fullstack Engineer— Jiffsy (Product)

February 2022 – May 2023

E-commerce acceleration platform • Shopify store optimization

Developed high-performance frontend solutions for Shopify stores, working with prominent Ukrainian brands including Zhilyova and Bevza to deliver exceptional mobilefirst shopping experiences.

- Built responsive, mobile-first UI components including virtualized lists, image sliders, and drawer navigation systems optimized for e-commerce conversion
- Developed proxy backend services to enhance Shopify store performance and functionality
- Implemented complex state management and data flow for real-time inventory and cart synchronization
- Collaborated with design and product teams to deliver pixel-perfect implementations of brand-specific shopping experiences
- Mentored junior developers while contributing to UI component library development
- Optimized application performance for mobile devices, ensuring smooth user experience across various screen sizes

Frontend Engineer— PVBKI (First Ukrainian Credit Bureau)

September 2021 – May 2022

Credit reporting and risk assessment • Financial services

Served as the sole frontend developer, creating internal tools and customer-facing applications for credit history management and risk assessment in the Ukrainian financial sector.

- Developed comprehensive credit history management application using Next.js with React Hook Form and Tailwind CSS, implementing full CRUD operations for financial data
- Built personality assessment questionnaire application using React and Chakra UI with Webpack compilation, designed to help financial institutions minimize lending risks
- Integrated with ASP.NET Core MVC backend services for secure financial data processing
- Implemented responsive design patterns ensuring accessibility across desktop and mobile platforms

Frontend Engineer— Sieve (Product)

February 2021 – February 2022

Content management platform • Personal project

Co-founded and developed a comprehensive content aggregation platform, building the entire frontend application and Chrome extension from the ground up.

- Architected and developed complete React.js frontend application using Next.js and Tailwind CSS
- Created Chrome extension available on Chrome Web Store, integrating with Chrome API for seamless content collection
- Implemented REST API integration for content synchronization and management
- Designed responsive user interface with styled-components for optimal user experience across devices

Languages

- English – Upper-Intermediate (daily working language)
- Ukrainian – Native